

Program 3

Code a Dockerized Python Flask or [Node.js](#) Application

Description

The problem statement focuses on creating a simple web application using either Python flask or [Node.js](#) and deploying it inside a Docker container. This ensures environment consistency and portability across different platforms.

Project Structure

Program3/
→ Dockerfile
→ requirements.txt
→ [app.py](#)

Step 1: Create the Dockerfile and add the similar contents

```
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8:~/Devopslab/program3$ cat Dockerfile
FROM python:latest

WORKDIR /app

COPY requirements.txt .
RUN pip install -r requirements.txt

COPY . .

EXPOSE 5000

CMD [ "python", "app.py"]
```

Step 2: Create a file → [app.py](#)

```
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8:~/Devopslab/program3$ cat app.py
from flask import Flask

app=Flask(__name__)

@app.route('/')

def home():
    return "hello from simple flask docker!"

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

Step 3: Make a file → requirements.txt

```
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8:~/DevopsLab/program3$ cat requirements.txt
Flask==2.3.3
```

Step 4: Execute the docker build command

```
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8:~/DevopsLab/program3$ docker build -t program3 .
[+] Building 34.0s (10/10) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 183B
=> [internal] load metadata for docker.io/library/python:latest
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [1/5] FROM docker.io/library/python:latest@sha256:1ad1a43b5e2478e62056bbc28020afd858185d73bf4d6a439cbb058b6800a96d
=> [internal] load build context
=> => transferring context: 93B
=> CACHED [2/5] WORKDIR /app
=> CACHED [3/5] COPY requirements.txt .
=> [4/5] RUN pip install -r requirements.txt
=> [5/5] COPY . .
=> exporting to image
=> => exporting layers
=> => writing image sha256:01af10a6844439c690acc6de0e4ad2e8f38d6120bab889ebb99af4bd91852187
=> => naming to docker.io/library/program3
```

docker:default
0.0s
0.0s
4.3s
0.0s
0.0s
0.0s
0.0s
0.0s
0.0s
0.0s
0.0s
0.0s
29.3s
0.0s
0.2s
0.2s
0.0s
0.0s

Step 5: Run the docker run command

```
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8:~/DevopsLab/program3$ docker run -it -p 3000:3000 program3
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.17.0.2:5000
Press CTRL+C to quit
```

Step 6: Check <http://localhost:5000> you will see the output that you want

